GAGANA B S

Tel: 7795059004 | gaganasuma61@gmail.com | Shivamogga, India |
www.linkedin.com/in/gagana-b-s-a5807737a

PROFESSIONAL SUMMARY

Currently a final-year Electrical and Electronics Engineering student with a strong foundation in Embedded Systems, C programming, Digital Electronics, Microcontrollers, and basic communication protocols. Hands-on experience in developing academic projects involving ESP32, Arduino, sensors, IoT, and wireless power transfer. Interested in power systems, power electronics, Embedded Systems, Firmware development and Electronics. With a strong willingness to learn and apply technical skills to real-world engineering problems. Seeking an opportunity to apply academic knowledge in a real-world environment, contribute to team success, and gain hands-on industry experience.

EDUCATION

Bachelor of Engineering in Electrical and Electronics (pursuing)

Jawaharlal Nehru New College of Engineering, VTU University CGPA: **8.51/10.0**

Higher secondary board of education

PUC in Mahesh PU college, Shivamogga with the **88%** in 2023

Secondary board of education

SSLC in San Thom English Medium school, Shivamogga with the **93%** in 2021

TECHNICAL SKILLS

- **Microcontrollers** (Arduino UNO, ESP32, STM32, MSP430) **–** Intermediate
- **Embedded C & C++** (timers, GPIO, interrupts – learning & applying)
- **Sensor interfacing, Communication protocols:** UART, SPI I2C
- **Soft Skills**: Problem Solving, Analytical Thinking, Teamwork, Communication, Adaptability, Quick Learning
- **Engineering Tools:** MATLAB/Simulink, STM32CubeIDE, Arduino IDE, VS Code, Thonny, Github (intermediate)

PROJECTS

TO CREATE A TCP SERVER FOR THE CLIENT USING ARDUINO

Establishes real-time communication between Arduino and client devices using TCP protocol over Wi-Fi/Ethernet. Enables remote monitoring and control of hardware components by exchanging data between client and Arduino server.

SECURITY SURVEILLANCE ALARM (OBJECT DETECTOR) USING ARDUINO UNO, SERVOMOTOR, LCD DISPLAY AND SENSORS

Detects objects and triggers an alert using sensors, while a servo motor can rotate to scan or track the detected object. Displays real-time status on LCD and activates an alarm system for security monitoring.

SOLAR BASED WIRELESS POWER TRANSFER ON ROAD FOR ELECTRIC VEHICLE

Developed a solar-powered wireless power transfer prototype for electric vehicles using inductive coupling to demonstrate contactless energy transmission and charging. Integrated Arduino-based monitoring and control with sensors and display modules to detect vehicle presence and indicate real-time charging status on the system

A MACHINE LEARNING APPROACH TO CONTROL A PROSTHETIC ARM FOR RESIDUAL LIMB – A BOON FOR AMPUTEES *(Present - final year project)*

CERTIFICATIONS

- NPTEL online course in enhancing soft skills and personality development
- NPTEL online course in the joy of computing using python
- Certificate in INFOYSYS SPRINGBOARD- basics of python
- L& T EDUTECH - Fundamentals of electric and hybrid electric vehicle technology
- L& T EDUTECH - EV charging infrastructure, vehicle testing & homologation
- L& T EDUTECH - Electric drive systems: batteries, powertrains and transmissions
- L& T EDUTECH – Embedded systems for electric vehicle

ACADEMIC ACHIEVEMENTS & ACTIVITIES

- Co-ordinated department technical event – TECHSPARK 1.0 & 2.0
- Participated in 24 HOUR HARDWARE HACKATHON
- Successfully completed a 6-day training program at NAANDI FOUNDATION

LANGUAGES

Kannada, English, Hindi